Multithreading 2-Day Master Plan

Multithreading 2-Day Master Plan (Amazon SDE Focus)

DAY 1 — Core Foundations (Executors → Synchronizers)

1. Executors

- ExecutorService, ThreadPoolExecutor
- Fixed, Cached, Scheduled thread pools
- submit() vs execute()
- Shutdown vs shutdownNow

Practice:

- Create fixed pool (3 threads)
- Submit 10 tasks
- Observe scheduling

2. Async Concepts

- Future, Callable, CompletableFuture

Practice:

- supplyAsync → thenApply → thenAccept
- allOf() for parallel tasks

3. Locks

- ReentrantLock, ReadWriteLock, Condition

Practice:

- Use lock.lock()/unlock()

- Producer–Consumer using Condition

4. Synchronizers

- CountDownLatch, CyclicBarrier, Semaphore, Phaser

Practice:

- CountDownLatch for 3 services
- Semaphore for limited access
- CyclicBarrier for 3-thread barrier

DAY 2 — Advanced Tools (Collections → Atomic → ForkJoin → Scheduler)

5. Concurrent Collections

- ConcurrentHashMap, BlockingQueue, CopyOnWriteArrayList

Practice:

- Producer/Consumer using BlockingQueue
- computeIfAbsent on CHM

6. Atomic Classes

- AtomicInteger, AtomicLong, CAS

Practice:

- High concurrency counter

7. Parallelism

- ForkJoinPool, RecursiveTask, Parallel Streams

Practice:

- Merge sort using ForkJoin

8. Scheduling

- ScheduledExecutorService

Practice:

- scheduleAtFixedRate vs scheduleWithFixedDelay

Amazon Interview Revision

- Thread vs Runnable vs Callable
- Deadlock, Livelock, Starvation
- volatile vs synchronized
- Atomic vs Lock
- CompletableFuture vs Future